Testing and Estimating Causal Heterogeneous Treatment Effects with Revised Deep Learning Semiparametric Regression in Observational Studies

Abstract

Estimating heterogeneous treatment effects (HTE) is pivotal for precision medicine and other domains where individualized decision-making is essential. In lung transplantation, for example, the benefit of bilateral-lung transplantation (BLT) versus single-lung transplantation (SLT) can differ markedly across recipients and depends on donor and recipient characteristics, including age, sex, lung allocation score (LAS), comorbidity burden, and baseline functional status. However, uncovering clinically meaningful, covariate-dependent treatment effect heterogeneity from observational data is challenging due to high-dimensional covariates and complex confounding structures. To address this, we propose *deepHTL*, a revised weighted deep learning semiparametric regression framework that both (i) robustly estimates HTE under complex confounding and (ii) provides a valid global test for the existence of HTE with controlled Type I error. Our extensive simulation studies show that *deepHTL* outperforms existing competing approaches, achieving lower estimation error while consistently maintaining nominal Type I error across a wide range of scenarios, including those with strong and nonlinear confounding. Applied to a large-scale lung-transplant cohort, *deepHTL* reveals significant HTE between SLT and BLT: younger, lower-risk recipients with low LAS, low BMI, and favorable functional profiles derive the greatest functional gains from BLT, directly informing patient selection, organ allocation, and surgical planning. An R package implementing the proposed *deepHTL* has been developed and is available on GitHub, facilitating its application in routine clinical and health-services research settings.

1. Introduction

Individualized decision-making has become essential in modern biomedical research and clinical care, particularly in precision medicine, risk stratification, and resource-constrained treatment allocation. Clinicians increasingly need to know not only whether a therapy is effective on average, but *for whom* it is beneficial, neutral, or harmful. Causal heterogeneous treatment effects (HTE) provide precisely this information: they characterize how the causal effect of a treatment varies across patients with different demographic

profiles, comorbidity burdens, biomarkers, and social determinants of health. In practice, ignoring HTE can lead to overtreatment of patients who are unlikely to benefit (and who may experience avoidable toxicity or complications) and undertreatment of patients for whom the same intervention is life-saving. For instance, in oncology, targeted therapies or immunotherapies may dramatically improve survival in biomarker-defined subgroups while offering little benefit and substantial toxicity in others; in cardiology, aggressive revascularization or implantable cardioverter-defibrillators may be warranted for carefully selected high-risk patients but confer marginal gain for those at lower baseline risk. A similar pattern appears in lung transplantation, where the clinical benefit of bilateral lung transplantation (BLT) compared with single-lung transplantation (SLT) can differ markedly across recipients: age, BMI, baseline functional status, frailty, and comorbidity burden all influence postoperative recovery and graft performance, leading to substantial covariate-dependent variation in BLT versus SLT outcomes [1]. Estimating HTE in these settings enables risk–benefit tailoring of therapies, refinement of clinical guidelines (e.g., defining which patients should receive an intensive or high-cost intervention), and more equitable allocation of scarce resources such as donor organs, ICU beds, or expensive biologics [2]. Beyond health care, HTE also play a critical role in policy evaluation, where the impact of job-training programs, conditional cash transfers, or geographically targeted interventions often varies systematically across socioeconomic strata [3, 4].

However, accurately recovering *causal* HTE from observational data is highly challenging. Unlike randomized controlled trials (RCTs), where treatment assignment is independent of potential outcomes by design, treatments in observational studies are often selected based on patient characteristics, disease severity, clinician preference, or health-system constraints. This nonrandom assignment introduces confounding, and treatment effect estimates can be severely biased if these confounding factors are not appropriately adjusted for [5]. The problem becomes even more acute when the goal is to characterize *how* effects vary across patients rather than to report a single average effect. In high-dimensional settings, many covariates including clinical risk scores, laboratory measurements, imaging-derived features, and socioeconomic indicators, may jointly influence both treatment choice and outcome through complex, nonlinear relationships and interactions. Even when standard identifiability conditions (e.g., no unmeasured confounding and overlap) are satisfied, accurately estimating nuisance functions (e.g., propensity scores and outcome regressions) in such settings is difficult, and errors in these nuisance estimates can propagate into biased, unstable, or spurious HTE estimates. From a practical standpoint, it is often not clear *a priori* whether clinically meaningful heterogeneity in treatment effect truly exists or whether observed differences reflect random variation and model overfitting. In many biomedical applications, the primary question is first to determine whether treatment effect heterogeneity is present at all, and only then to characterize its pattern if it exists. Without a rigorous global test for the existence of HTE, investigators may over-interpret noisy subgroup effects, inflate Type I error through

multiple subgroup searches, or deploy unnecessarily complex individualized treatment rules when a simpler "one-size-fits-most" strategy would suffice. Yet formal global tests for the presence of HTE in observational studies with valid Type I error control under complex confounding and high-dimensional covariate structures remain underdeveloped.

An extensive literature in causal machine learning has proposed flexible estimators of HTE. For example, meta-learning strategies (e.g., T-learner, S-learner, X-learner) [6] leverage off-the-shelf prediction algorithms for each treatment arm or for transformed outcomes. Bayesian causal forests and related tree-based methods [7, 8] offer powerful nonparametric modeling of individualized effects and uncertainty quantification, but can be sensitive to prior specification, hyperparameter tuning, and MCMC convergence, especially in high-dimensional spaces. Counterfactual generative methods such as GANITE [9] attempt to learn the joint distribution of treated and untreated potential outcomes via adversarial training, but their performance can deteriorate when overlap is limited or when treatment selection involves high-order interactions among confounding covariates. Importantly, most of these approaches primarily focus on improving the accuracy of point or interval estimation of HTE, without providing a calibrated global test for the existence of treatment effect variation. Existing heterogeneity tests either target randomized trials [10] or examine pre-specified subgroup differences in observational settings [11], and do not readily extend to real-world observational data with complex confounding structures. Semiparametric regression models based on Robinson's transformation, which effectively cancels out complex nuisance confounding effects, provide an appealing alternative for causal HTE estimation. The quasi-oracle R-learner proposed by [12] recasts the problem as a residual-on-residual regression, yielding orthogonal (or Neyman-orthogonal) estimating equations that mitigate first-order bias from nuisance estimation. In principle, this framework can be combined with flexible machine learning models for both nuisance functions and the treatment effect function. However, in practical observational studies, e.g., large organ transplantation registries, electronic health record cohorts, or biobank-based studies, standard nuisance learners (e.g., logistic regression, kernel ridge regression, XGBoost) may either underfit complex nonlinear confounding structures or overfit in finite sample settings, compromising the stability of the R-learner. Additionally, when the underlying treatment effect is large in magnitude or the random noise variance is small, the continuous nuisance conditional expectation can be dominated by the binary nuisance conditional expectation, making it difficult to obtain accurate nuisance estimates and reliable treatment effect inference in finite samples [13]. Current deep learning semiparametric regressions partially address these issues but typically focus on improving HTE estimation accuracy, without providing an integrated framework that both delivers efficient HTE estimation and supports rigorous statistical inference on the existence of HTE.

Motivated by these gaps and by the need to support high-stakes clinical decision-making, we develop

a revised deep learning semiparametric regression framework, *Deep Heterogeneous Treatment Learning* (*deepHTL*), for testing and estimating causal HTE in observational studies. Our approach couples bagged deep neural networks (bagged-DNN) for robust nuisance function estimation with an orthogonalized residual-on-residual weighted regression and a bias-correction revision designed to reduce bias in estimating the nuisance conditional expectations under finite sample settings. To effectively assess whether treatment effect variation exists, we construct a Davies-type global test based on orthogonalized residual scores and a kernelized covariance structure, and we complement this analytic test with a permutation-based benchmark that empirically evaluates Type I error control under various confounding structures. Together, these components yield a unified framework (*deepHTL*) that both tests for and estimates causal heterogeneous treatment effects under realistic conditions encountered in clinical practice. We evaluate the proposed method through extensive simulations, assessing Type I error control of the analytical (Davies test) and empirical (permutation test) methods and comparing the estimation performance of *deepHTL* against the R-learner across a range of confounding complexity scenarios. We then apply the framework to a large United Network for Organ Sharing (UNOS) lung-transplant cohort to investigate how transplant type, i.e., BLT versus SLT, affects post-transplant lung function, measured by forced expiratory volume in one second (FEV1). The analysis reveals clinically meaningful patterns: younger and lower-risk recipients show the greatest functional gains from BLT, illustrating how causal HTE estimation can inform patient-specific decisions for a scarce and high-risk intervention. While our application focuses on lung transplantation, the proposed methodology readily generalizes to other observational studies where individualized treatment effects are of primary interest.

The remainder of the paper is organized as follows. Section 2 presents the proposed methodological framework, Section 3 reports extensive simulation results comparing the proposed framework with existing competing methods, Section 4 provides the lung-transplant application using our proposed method, and Section 5 concludes with a brief discussion.

2. Methods

To model potentially complex heterogeneous treatment effects and accommodate intricate confounding structures common in practical biomedical and clinical studies, we consider the following semiparametric model:

$$Y_i = \tau(\boldsymbol{X}_i)Z_i + f(\boldsymbol{X}_i) + \varepsilon_i$$

where $Y_i, Z_i, \boldsymbol{X}_i$ denote the observed outcome, treatment assignment, and covariates for the ith individual ($i = 1, \ldots, n$). The random error term ε_i satisfies $\mathbb{E}[\varepsilon_i \mid \boldsymbol{X}_i, Z_i] = 0$ and $\mathrm{Var}[\varepsilon_i \mid \boldsymbol{X}_i, Z_i] = \sigma^2$. To robustly

capture complex confounding effects on the outcome, we leave the nuisance function $f(\boldsymbol{X})$ unspecified. Furthermore, we define the treatment effect of interest, $\tau(\boldsymbol{X})$, as an unspecified function of the covariates $\boldsymbol{X}$, allowing for flexible modeling of heterogeneity. Under this semiparametric framework, we present a robust and efficient method to estimate the HTE $\tau(\boldsymbol{X})$ and test for its existence, as described below.

2.1. Deep Heterogeneous Treatment Learning for Estimating HTE

To avoid estimating the unknown nuisance function $f(\boldsymbol{X})$ in the above semi-parametric model, we employ the Robinson transformation:

$$Y - \mu(\boldsymbol{X}) = \tau(\boldsymbol{X})\{Z - e(\boldsymbol{X})\} + \varepsilon, \tag{2.1}$$

where $\mu(\boldsymbol{X}) = \mathbb{E}[Y \mid \boldsymbol{X}]$ and $e(\boldsymbol{X}) = \mathbb{E}[Z \mid \boldsymbol{X}]$ represent nuisance functions that require accurate pre-estimations [14]. Standard machine learning methods such as Lasso, XGBoost, and Kernel Ridge Regression, as used in [12], often struggle when the dimension of covariate space is high and the confounding structure is complex. In such settings, nuisance estimators trained on finite samples can exhibit systematic underfitting, and such errors would accumulate and propagate into the estimation of $\tau(\boldsymbol{X})$. To obtain reliable nuisance estimates, we implement a bagged-DNN model enhanced with a filtering mechanism [13]. This ensemble aggregates predictions from multiple bootstrap-trained deep neural networks while explicitly discarding poorly converged models, thereby stabilizing the fits against the randomness of network initialization. To further mitigate overfitting, we adopt a K-fold cross-fitting scheme in which, for each fold, the bagged-DNN models for $\mu(\boldsymbol{X})$ and $e(\boldsymbol{X})$ are trained on the training folds and used to generate predictions exclusively for the held-out observations. Using these nuisance estimates $\hat{\mu}(\boldsymbol{X})$ and $\hat{e}(\boldsymbol{X})$, we obtain a preliminary estimator of heterogeneous treatment effect by minimizing the weighted least squares loss based on (2.1):

$$\hat{\tau}(\boldsymbol{X}_i) = \arg\min_{\tau} \frac{1}{n} \sum_{i=1}^{n} w_i \Big\{ Y_i - \hat{\mu}(\boldsymbol{X}_i) - \tau(\boldsymbol{X}_i)\,\{Z_i - \hat{e}(\boldsymbol{X}_i)\} \Big\}^2, \tag{2.2}$$

where the weights are defined as $w_i = (Z_i - \hat{e}(\boldsymbol{X}_i))^2$, and the loss is minimized via a bagged-DNN [13]. Crucially, this objective function satisfies the property of Neyman orthogonality, ensuring that the impact of first-order estimation errors in the nuisance functions is asymptotically negligible [15]. However, $\hat{\tau}(\boldsymbol{X})$ can be severely biased when the magnitude of $\tau(\boldsymbol{X})$ is large. As shown in (2.1), the residual $Y - \mu(\boldsymbol{X})$ comprises a binary component $\tau(\boldsymbol{X})\{Z - e(\boldsymbol{X})\}$ and a random noise ε. When $|\tau(\boldsymbol{X})|$ is small, the residual is dominated by ε, which is typically well approximated by a unimodal Gaussian error. In this context, minimizing the squared error in estimating $\mu(\boldsymbol{X})$ is effectively equivalent to maximizing the Gaussian log-likelihood and is therefore statistically efficient. Conversely, when $|\tau(\boldsymbol{X})|$ is large, the binary component dominates and the

Algorithm 1 deepHTL: Deep Heterogeneous Treatment Learning

1: **Input**: Observations $\{(Y_i, Z_i, \boldsymbol{X}_i)\}_{i=1}^n$
2: Estimate nuisance functions $\hat{e}(\boldsymbol{X})$ and $\hat{\mu}(\boldsymbol{X})$ using bagged-DNN.
3: Compute the constant effect $\hat{\tau}_0$ using (2.4) and the centered outcome $Y^* = Y - \hat{\tau}_0 Z$.
4: Estimate the centered nuisance function $\hat{\mu}^*(\boldsymbol{X}) = \mathbb{E}[Y^* \mid \boldsymbol{X}]$ via bagged-DNN.
5: Estimate the residual heterogeneity $\hat{r}(\boldsymbol{X})$ via weighted least squares of $Y^* - \hat{\mu}^*(\boldsymbol{X})$ on $Z - \hat{e}(\boldsymbol{X})$.
6: **Output**: Revised HTE estimate $\hat{\tau}^*(\boldsymbol{X}) = \hat{\tau}_0 + \hat{r}(\boldsymbol{X})$.

residual distribution becomes markedly bimodal. Under these conditions, squared-error estimation of $\mu(\boldsymbol{X})$ becomes substantially less efficient.

To address this issue, we decompose the treatment effect into a constant component and a residual heterogeneous component: $\tau(X) = \tau_0 + r(X)$. Under this decomposition, we reformulate (2.1) by replacing the Robinson equation (2.2) with the following modified model:

$$Y^* - \mathbb{E}[Y^* \mid \boldsymbol{X}] = r(\boldsymbol{X}) \left\{ Z - e(\boldsymbol{X}) \right\} + \varepsilon, \tag{2.3}$$

where $Y^* = Y - \tau_0 Z$ denotes the centered outcome and $\mu^*(\boldsymbol{X}) = \mathbb{E}(Y^* \mid \boldsymbol{X})$ is estimated using the bagged-DNN. This revision substantially reduces the dominant binary component and remarkably strengthens the unimodal residual structure, allowing $\mu^*(\boldsymbol{X})$ to be estimated more accurately. In practice, many analytical methods can be adopted to pre-estimate τ_0. Here, we set τ_0 to an estimate of the average treatment effect using a simple propensity score adjustment via the estimated $\hat{e}(\boldsymbol{X})$ which is computationally efficient and straightforward and intuitively provides a natural centering of the overall treatment effect:

$$\hat{\tau}_0 = \frac{\sum_{i=1}^n \{Y_i - \hat{\mu}(\boldsymbol{X}_i)\}\{Z_i - \hat{e}(\boldsymbol{X}_i)\}}{\sum_{i=1}^n \{Z_i - \hat{e}(\boldsymbol{X}_i)\}^2}. \tag{2.4}$$

The residual heterogeneous component estimator $\hat{r}(\boldsymbol{X})$ is then obtained by solving the weighted least squares problem:

$$\hat{r}(\boldsymbol{X}_i) = \arg\min_r \frac{1}{n} \sum_{i=1}^n w_i \Big\{ Y_i - \tau_0 Z_i - \hat{\mu}^*(\boldsymbol{X}_i) - r(\boldsymbol{X}_i) \left\{ Z_i - \hat{e}(\boldsymbol{X}_i) \right\} \Big\}^2.$$

Combining the constant and residual components produces the revised estimator of the heterogeneous treatment effect:

$$\hat{\tau}^*(\boldsymbol{X}) = \hat{\tau}_0 + \hat{r}(\boldsymbol{X}). \tag{2.5}$$

A complete summary of the estimation procedure is provided in Algorithm 1.

2.2. Testing for the Existence of HTE

In clinical practice, it is critically important to test the existence of HTE to avoid deploying unnecessarily complex individualized treatment rules. In the absence of heterogeneity, the average treatment effect (ATE) provides a sufficient summary of the treatment effect, and the computational cost of bagged-DNN modeling can be avoided. We therefore test the null hypothesis $H_0 : \mathrm{Var}\{\tau(\boldsymbol{X})\} = 0$, which implies that $\tau(\boldsymbol{X})$ reduces to an unknown constant τ. Our testing framework integrates an analytic Davies test [16] with an empirical permutation procedure that provides a robust finite-sample benchmark.

Studentized Residual Scores. In practice, true treatment effect τ can be large in magnitude, and the construction of a valid test requires an accurate estimate of the constant (average) treatment effect under H_0. The preliminary estimator $\hat{\tau}_0$ from (2.4) inherits bias when the magnitude of τ is large. We therefore adopt a revised null estimator by substituting τ for $\tau(\boldsymbol{X})$ in (2.3):

$$\hat{\tau}^* = \hat{\tau}_0 + \frac{\sum_{i=1}^n \{(Y_i - \hat{\tau}_0 Z_i - \hat{\mu}^*(\boldsymbol{X}_i)\}\{(Z_i - \hat{e}(\boldsymbol{X}_i)\}}{\sum_{i=1}^n \{Z_i - \hat{e}(\boldsymbol{X}_i)\}^2}. \tag{2.6}$$

To construct the test statistic, we derive residuals that quantify deviations from the null model. Incorporating the revised estimator $\hat{\tau}^*$, we define the residual score for the i-th observation as: $r_i = \frac{Y_i - \hat{\tau}_0 Z_i - \hat{\mu}^*(\boldsymbol{X}_i)}{Z_i - \hat{e}(\boldsymbol{X}_i)} - (\hat{\tau}^* - \hat{\tau}_0)$, which corresponds to the estimated regression residual. Under H_0, Neyman orthogonality ensures that the impact of nuisance estimation errors is asymptotically negligible, so that $r_i \approx \varepsilon_i / \{Z_i - e(\boldsymbol{X}_i)\}$. To ensure unit variance, we compute the studentized residual $s_i = \sqrt{w_i}\, r_i / \hat{\sigma}$, where $w_i = \{Z_i - \hat{e}(\boldsymbol{X}_i)\}^2$ and $\hat{\sigma}^2 = n^{-1} \sum_{i=1}^n w_i r_i^2$ provides an estimate of $\mathrm{Var}(\varepsilon_i | \boldsymbol{X}_i, Z_i)$.

Davies Test of HTE. Let $s = (s_1, \ldots, s_n)^\top$ and $\sqrt{W} = (\sqrt{w_1}, \ldots, \sqrt{w_n})^\top$ denote the n-dimensional vectors of studentized residuals and weight roots, respectively. The estimation of $\hat{\tau}^*$ imposes a linear constraint on these scores. Specifically, the weighted least squares estimating equation enforces the orthogonality condition $\sqrt{W}^\top s = 0$, which consumes one degree of freedom and induces a rank-one dependency in the covariance structure of s. As a result, the covariance structure of s becomes singular and no longer coincides with the identity matrix I_n, invalidating classical reference distributions that assume independent errors. To account for this induced dependence, we project the residual scores onto the orthogonal complement of the subspace spanned by $\sqrt{W}$ using the projection matrix

$$P = I_n - \frac{\sqrt{W}\,\sqrt{W}^\top}{\sum_{i=1}^n W_i}.$$

We then construct the test statistic that detects structured departures from the null model:

$$Q = (Ps)^\top K(Ps),$$

where we use a Gaussian kernel K to flexibly accommodate nonlinear structure in the treatment effect while avoiding reliance on restrictive parametric assumptions [17]. Under H_0, the test statistic Q asymptotically follows a weighted mixture of chi-square distributions. We compute the analytic p-value using Davies method [16], which evaluates the tail probability of this mixture distribution via inversion of its characteristic function.

Permutation Test of HTE. While the analytic Davies test is computationally efficient, its accuracy relies on asymptotic approximations that may degrade in finite samples. To provide a robust benchmark, we implement a permutation test that generates a valid reference distribution without imposing distributional assumptions. Keeping the projection matrix P and kernel K fixed, we randomly permute the studentized residual s a total of B times and recompute the quadratic-form statistics

$$Q^{(b)} = (Ps^{(b)})^\top K(Ps^{(b)}), \qquad b = 1, \ldots, B,$$

which approximates the sampling distribution of the test statistic under H_0. The permutation p-value is computed as

$$p^{\text{perm}} = \frac{1 + \sum_{b=1}^{B} \boldsymbol{X} 1\{Q^{(b)} \geq Q\}}{B+1}.$$

3. Simulation Studies

We conduct extensive simulations under various data complexity settings to mimic the practical data structures to evaluate both (i) the finite-sample performance of the proposed testing methods under the null hypothesis $H_0 : \text{Var}\{\tau(\boldsymbol{X})\} = 0$ and (ii) the predictive performance of the proposed HTE estimators under various confounding settings. Throughout, we adopt designs that deliberately induce complexity in the nuisance functions to stress-test the proposed framework. Specifically, data are generated according to $\boldsymbol{X} \sim \mathcal{N}_p(0, I_p)$, $Z \mid \boldsymbol{X} \sim \text{Bernoulli}(e(\boldsymbol{X}))$, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, with the outcome model specified as

$$Y \;=\; b(\boldsymbol{X}) + \tau(\boldsymbol{X})\big(Z - e(\boldsymbol{X})\big) \;+ \varepsilon.$$

The baseline outcome function incorporates complex, nonlinear interactions, $b(\boldsymbol{X}) = \log(|\boldsymbol{X}_1| + 1) - \boldsymbol{X}_2^2 + \sin(\boldsymbol{X}_3) + 0.5\,\boldsymbol{X}_4\boldsymbol{X}_5$. The propensity score is likewise driven by higher-order interactions, $e(\boldsymbol{X}) =$

$\text{logit}^{-1}\{0.8\sin(\pi \boldsymbol{X}_1\boldsymbol{X}_2) + 0.6\boldsymbol{X}_3\boldsymbol{X}_4 + 0.5\tanh(\boldsymbol{X}_5)\}$. To assess performance across regimes of sample size and complexity, we consider a grid of $n \in \{1000, 2000\}$, covariate dimensions $p \in \{20, 40\}$, and noise levels $\sigma \in \{1, 3\}$. For each configuration, we generate 1000 independent replicates. The specification of $\tau(\boldsymbol{X})$ varies by subsection to match the corresponding evaluative goal.

3.1. Performance on Controlling Type I Error

To assess finite-sample validity, we evaluate type I error control under a constant treatment effect. Within the general design described above, we set the treatment effect to be constant, considering both a small effect $\tau = 0$ and a large-magnitude effect $\tau = 3$. We implement the test using both the preliminary estimator $\hat{\tau}_0$ from (2.4) and the revised estimator $\hat{\tau}_0^*$ from (2.6). Performance is measured by the empirical type I error rate $\hat{\alpha}$ at level $\alpha = 0.05$ for both the Davies test ($\hat{\alpha}^{\text{Davies}}$) and the permutation benchmark ($\hat{\alpha}^{\text{Perm}}$).

The results in Table 1 reveal clear patterns depending on the magnitude of the treatment effect. Under the small treatment effect ($\tau = 0$), both the preliminary and revised estimators achieve near-nominal type I error across most configurations, with only mild inflation (~ 0.08) in the most challenging setting ($n = 1000, p = 40, \sigma = 3$), where high dimensionality, limited sample size, and elevated noise hinder nuisance estimation. However, when a large treatment effect is present ($\tau = 3$), the necessity of the revision becomes evident. As discussed in Section 2.2, the residual $Y - \mu(\boldsymbol{X})$ decomposes into a binary component $\tau\{Z - e(\boldsymbol{X})\}$ and noise ε. In low-noise settings ($\sigma = 1$), a large τ causes the binary term to dominate, creating a highly non-Gaussian residual that makes squared-error estimation of $\mu(\boldsymbol{X})$ inefficient. This distortion propagates to the preliminary estimator $\hat{\tau}_0$, which leads to severe type I error inflation—for instance, with $n = 2000$, $p = 20$, and $\sigma = 1$, the empirical type I error for $\hat{\tau}_0$ reaches nearly 0.30. In contrast, when noise is high ($\sigma = 3$), the residual is again dominated by ε, restoring an approximately unimodal structure and greatly mitigating the inflation. The revised estimator $\hat{\tau}_0^*$ resolves this problem in low-noise, large-effect settings and restores type I error control, as shown in the bottom panel of Table 1.

Notably, the mild inflation observed under the most challenging condition ($p = 40, \sigma = 3$) subsides as the sample size increases from 1000 to 2000, dropping to approximately 0.06. This trend indicates that the deep learning nuisance estimators benefit from larger sample sizes. Given that our subsequent real-world data analysis includes over 10,000 observations, we anticipate rigorous validity in practice. Furthermore, the permutation benchmark closely mirrors the analytic Davies type I error, confirming that the quadratic-form test statistic Q is well-calibrated under the mixture chi-square approximation. These findings underscore that while the preliminary estimator can be dangerously liberal under large-effect and low-noise conditions, the revised framework consistently provides robust inference under complex confounding structures.

Table 1: Empirical type I error rates ($\alpha = 0.05$) based on the Davies and Permutation tests.

τ	Estimator	n	p	σ	$\hat{\alpha}^{\text{Davies}}$	$\hat{\alpha}^{\text{Perm}}$
$\tau = 0$	Preliminary ($\hat{\tau}_0$)	1000	20	1	0.042	0.045
		1000	20	3	0.056	0.058
		1000	40	1	0.062	0.068
		1000	40	3	0.083	0.086
		2000	20	1	0.045	0.044
		2000	20	3	0.058	0.055
		2000	40	1	0.055	0.057
		2000	40	3	0.060	0.060
	Revised ($\hat{\tau}_0^*$)	1000	20	1	0.044	0.044
		1000	20	3	0.057	0.061
		1000	40	1	0.070	0.074
		1000	40	3	0.086	0.090
		2000	20	1	0.043	0.042
		2000	20	3	0.057	0.057
		2000	40	1	0.053	0.058
		2000	40	3	0.059	0.060
$\tau = 3$	Preliminary ($\hat{\tau}_0$)	1000	20	1	0.176	0.175
		1000	20	3	0.064	0.068
		1000	40	1	0.152	0.168
		1000	40	3	0.086	0.096
		2000	20	1	0.299	0.304
		2000	20	3	0.072	0.073
		2000	40	1	0.181	0.185
		2000	40	3	0.070	0.072
	Revised ($\hat{\tau}_0^*$)	1000	20	1	0.044	0.044
		1000	20	3	0.057	0.061
		1000	40	1	0.063	0.069
		1000	40	3	0.085	0.089
		2000	20	1	0.043	0.041
		2000	20	3	0.053	0.055
		2000	40	1	0.055	0.057
		2000	40	3	0.060	0.060

3.2. Performance on Estimating HTE

Having validated Type I error control, we next evaluate the estimation accuracy of heterogeneous treatment effects. We retain the nuisance functions $b(\boldsymbol{X})$ and $e(\boldsymbol{X})$ from the previous section, but replace the constant effect with covariate-dependent $\tau(\boldsymbol{X})$ to assess predictive performance. Two contrasting scenarios are considered to reflect different levels of structural complexity. **Scenario I** represents a mild, nearly linear effect, $\tau(\boldsymbol{X}) = -1 + 2\log(\exp(\sum_{j=1}^{5} \boldsymbol{X}_j) + 1)$, whereas **Scenario II** introduces strong nonlinear interactions, $\tau(\boldsymbol{X}) = -1 + \boldsymbol{X}_1\boldsymbol{X}_2 + \cos^2(\boldsymbol{X}_3) + \max(\boldsymbol{X}_4 - \boldsymbol{X}_5, 0)$. For each configuration in the design grid, models are trained on one dataset and evaluated on an independent test set of equal size to assess out-of-sample

predictive performance.

To benchmark the proposed *deepHTL* framework, we compare it with three machine learning methods: least absolute shrinkage and selection operator (*Lasso*), eXtreme Gradient Boosting (*XGBoost*), and kernel ridge regression (*KRR*). Each learner is used consistently throughout the estimation pipeline to obtain the nuisance functions $\mu(\boldsymbol{X})$ and $e(\boldsymbol{X})$, as well as the treatment effect function $\tau(\boldsymbol{X})$. To isolate the contribution of our revision step, we implement two versions of every learner: the preliminary estimator in (2.2) (denoted ***R–learner***, e.g., *R–Lasso*) and the proposed revised estimator in (2.5)(denoted ***Rev–learner***, e.g., *Rev–XGBoost*). Within this comparison framework, *deepHTL* corresponds to the revised estimator implemented with bagged DNNs, while its preliminary counterpart is denoted *R–DNN*.

All estimators are tuned carefully to ensure a fair comparison. For DNN-based methods, we use a bagged ensemble of 30 feedforward neural networks with ReLU activations, hidden layers $(128, 64, 32)$, and ℓ_1 regularization (10^{-3}), optimized via Adam with early stopping. The Lasso uses cross-validated ℓ_1 regularization via `glmnet` [18]. XGBoost is tuned via a random grid search over tree depth, subsampling ratios, shrinkage rates, and minimum child weights. KRR uses a Gaussian kernel with bandwidth selected via cross-validation around the median heuristic. Nuisance components for all learners are estimated via 5-fold cross-fitting. Computations are carried out in R [19], and a complete implementation is available in an open-source R package on GitHub.

The comparative results, summarized by the logarithmic mean squared error log(MSE) in Figure 1, reveal clear and consistent patterns in the performance of the proposed framework. The utility of the revision depends heavily on the base learner's capacity. For the Lasso estimator, the revision produces little improvement. This is expected, as the Lasso's linear additive assumption prevents it from accurately learning the nonlinear structures in $\mu(\boldsymbol{X})$, leaving little room for the revised formulation to offer additional gains. Conversely, for flexible learners like XGBoost and KRR, the revision results in noticeable improvements, particularly in the Scenario II where the median reduction in MSE is approximately 18% ($\Delta \log(\text{MSE}) \approx 0.18$). In Scenario I, where the treatment effect function is relatively smooth, the benefits of the revision are more subtle for most methods, though the revised DNN still shows discernible gains as the sample size increases to 2000. Crucially, the revision never materially degrades performance in these complexity settings. While theoretical intuition suggests that the additional nuisance estimation steps may introduce minor noise in scenarios where the treatment effect surface is relatively simple and of modest magnitude, we observe no such penalty here, indicating that the revision serves as a safe enhancement that improves estimation in complex scenarios without compromising accuracy in simpler ones.

Most importantly, *deepHTL* outperforms all competing methods across the entire simulation grid, with the largest improvements observed at $n = 2000$ and $\sigma = 1$. These results align with our methodological motivation

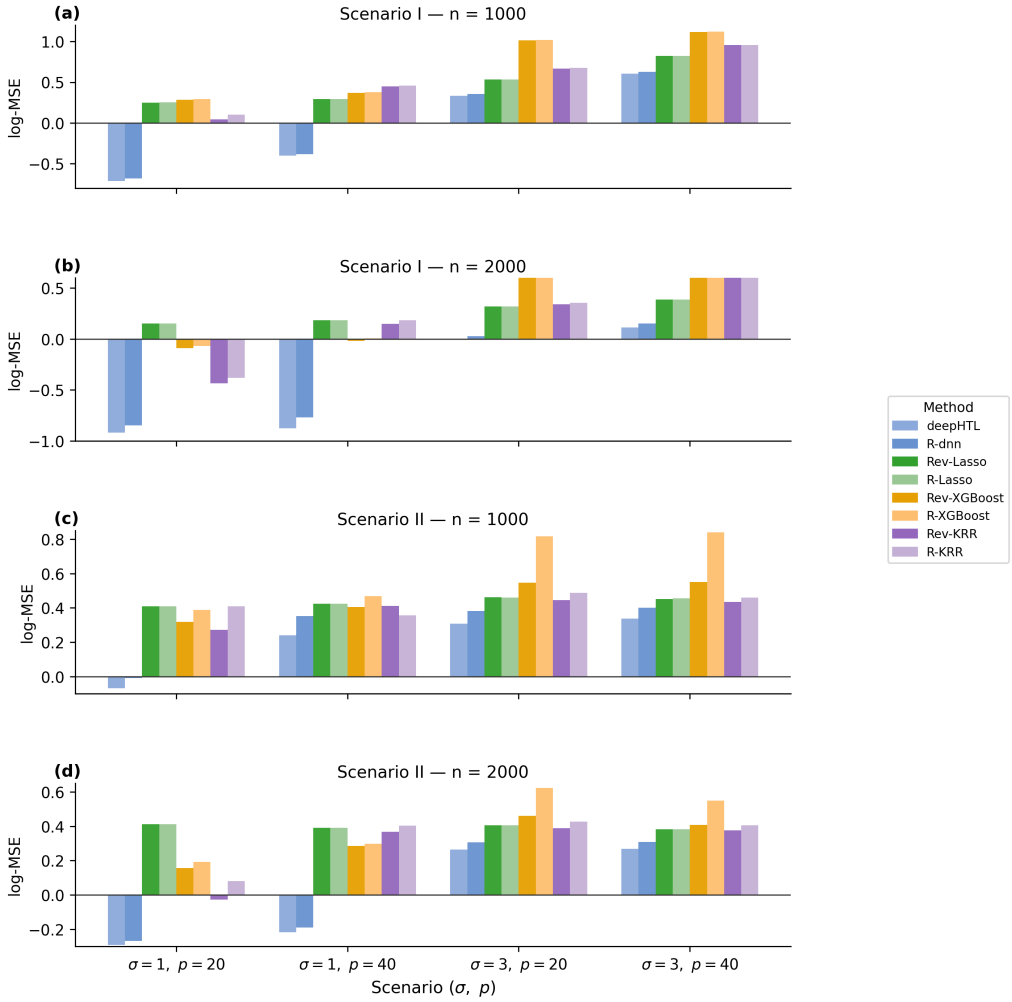

Figure 1: Mean log(MSE) of $\hat{\tau}(X)$ across simulation settings for all HTE estimators.

such that the revision allows the high-capacity neural network to accurately estimate the nuisance functions and thereby capture intricate treatment effect structures. For practitioners, this suggests a context-dependent selection strategy: while *deepHTL* is most effective in settings with complex confounding, large-magnitude effects, and sufficiently large sample sizes, simpler revised learners such as *R–Lasso* may be adequate in smoother, lower-dimensional problems where computational efficiency is a priority. Motivated by these results, we next apply *deepHTL* to a real-world lung case study in the following section.

4. A Lung Transplant Study

To demonstrate the practical utility of our proposed framework, we apply the proposed framework to registry data from the United Network for Organ Sharing (UNOS) to investigate a central question in lung

transplantation: whether bilateral lung transplantation (BLT) confers superior graft function relative to single lung transplantation (SLT), and whether this advantage varies across patients or clinically relevant subgroups. Although BLT is generally associated with improved survival, the extent of heterogeneity in its advantage remains underexplored [20, 21]. Our study cohort includes 14,306 adult recipients transplanted between May 2005 and September 2011. The outcome Y is the post-transplant Forced Expiratory Volume in one second (FEV1), a key indicator of graft performance and long-term survival risk [22]. Treatment is encoded as $Z \in \{0, 1\}$, where $Z = 1$ denotes BLT ($n = 9{,}758$) and $Z = 0$ denotes SLT ($n = 4{,}548$). The covariate vector $\boldsymbol{X}$ comprises baseline donor and recipient characteristics used to adjust for confounding. These variables were selected by a data-adaptive Group Lasso–Doubly Robust (GL-DR) procedure [23], which retained 26 clinically relevant predictors including recipient age, BMI, and ischemic time, from an initial set of 29 (Table 2).

We begin by assessing the presence of global heterogeneity using the Davies test based on the revised estimator in (2.6). To better understand the nature of the detected heterogeneity, we summarize its complexity using the Satterthwaite approximation, which represents the chi-square mixture through an effective degrees of freedom k_{eff} [24]. Conceptually, k_{eff} reflects how many independent eigen-directions of the kernel contribute meaningfully to the test statistic: values near 1 correspond to a simple, low-dimensional form of heterogeneity, whereas larger values indicate a more intricate, high-dimensional pattern. In our analysis, the Davies test returns a p-value of 0.002, providing strong evidence for HTE. The estimated $k_{\text{eff}} \approx 137.5$ is large, suggesting that the heterogeneity arises from many covariate directions rather than a small subset of variables. As a finite-sample check, a permutation test with $B = 20{,}000$ replicates produces a closely similar p-value of 0.002, indicating that the analytic approximation is well-calibrated.

Given the detected heterogeneity, we next estimate $\tau(\boldsymbol{X})$, referred to hereafter as the individual treatment effect (ITE), using *deepHTL*. For a population-level reference, we also estimate the ATE using a doubly robust AIPW estimator [25]. The estimated ATE of FEV1 is 18.7 (95% CI: 16.5, 20.9) mL/sec, indicating that, on average, BLT provides superior graft function relative to SLT. The distribution of $\hat{\tau}(\boldsymbol{X})$ (Figure 2) is broad and approximately bimodal, ranging from -20.6 to 59.2 mL/sec. While the sample mean of the $\hat{\tau}(X)$ (17.9 mL/sec) aligns closely with the estimated ATE, the variability is substantial: approximately 93% of patients ($n = 13{,}210$) are predicted to experience a functional gain from BLT, whereas roughly 7% ($n = 986$) show a negative effect. These results suggest that, despite BLT being the generally superior procedure, a uniform treatment strategy may expose certain patients to the greater surgical burden of BLT without commensurate gains.

To translate the individual estimates into clinically interpretable patterns, we compute Group Average Treatment Effects (GATEs) for key subpopulations (Figure 3). The advantage of BLT is largest among

Table 2: Key patient and donor characteristics used as confounders by transplant type (BLT vs. SLT). The complete 26 covariates selected are shown in the Appendices.

Covariates	**Transplant type**	
	BLT (n = 9758)	**SLT** (n = 4548)
Recipient Age (years), Mean (SD)	53.4 (13.5)	63.5 (7.5)
Recipient BMI, Mean (SD)	24.8 (4.7)	26.3 (4.1)
Lung Allocation Score, Mean (SD)	40.8 (13.5)	40.3 (11.1)
Diabetes, n (%)	1999 (20.5%)	836 (18.4%)
Recipient Height (cm), Mean (SD)	170.0 (9.9)	170.6 (9.6)
Ventilator Use, n (%)	465 (4.8%)	71 (1.6%)
6-Minute Walk Distance, Mean (SD)	773.8 (426.1)	794.6 (411.4)
Donor Age (years), Mean (SD)	34.7 (14.0)	34.9 (13.8)
Donor BMI, Mean (SD)	26.2 (5.6)	26.2 (5.3)
CMV Serology Positive, n (%)	6093 (62.4%)	2793 (61.4%)
Donor Diabetes, n (%)	701 (7.2%)	315 (7.0%)
Ischemic Time (hours), Mean (SD)	5.6 (1.76)	4.3 (1.52)

patients with high physiological reserve, particularly younger recipients (<50 years), those with low BMI (<18.5), and those with lower acuity (LAS < 35). Conversely, the benefit diminishes markedly for older recipients ($\geq$75 years) and individuals with high metabolic burden (obese class II/III, BMI ≥ 30). Donor graft characteristics such as ischemic time exert comparatively smaller influence on differential treatment benefit. These findings suggest a principled resource-allocation strategy: limited bilateral grafts provide the greatest marginal improvement among younger, low-risk, and lower-BMI candidates, whereas SLT may remain an adequate option for older or higher-risk recipients. As a robustness check, an unsupervised k-means clustering of the estimated ITEs (Appendices) recovers clinically meaningful phenotypes that closely mirror the GATE patterns, indicating that the observed heterogeneity is driven by these core physiological characteristics.

Our findings should be interpreted in light of several limitations inherent to observational registry analyses. First, causal inference relies on the assumption of no unmeasured confounding; while we adjust for 26 key clinical covariates selected via a robust GL-DR procedure, unrecorded factors such as center-specific surgical protocols or granular frailty metrics could influence outcomes. Second, we assume missing data were ignorable conditional on observed covariates, which may not hold for all variables. Finally, since the study cohort spans $2005 - 2011$, generalizability to current practice may be affected by subsequent changes in donor allocation policies and surgical techniques. Despite these caveats, the analysis demonstrates the capacity of the proposed framework to uncover actionable heterogeneity in large-scale data.

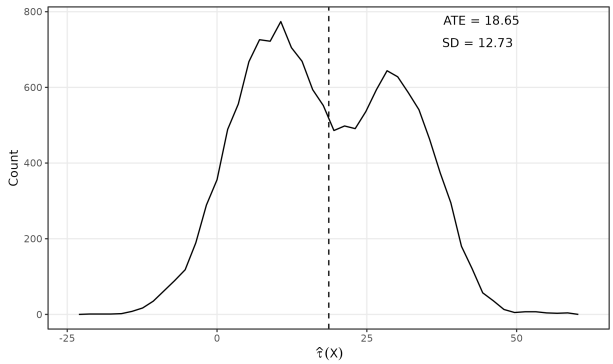

Figure 2: Estimated individualized treatment effects (BLT vs. SLT) in the whole UNOS lung transplant cohort. The dashed line denotes the average treatment effect.

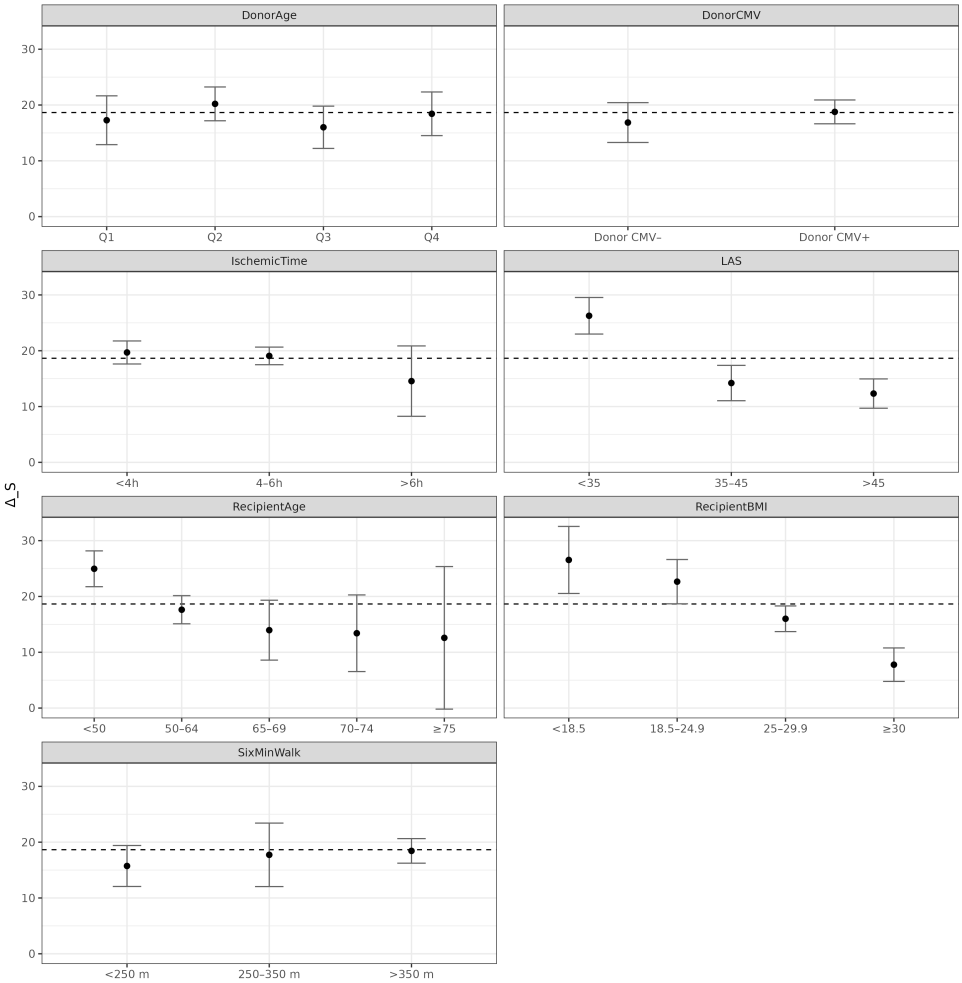

Figure 3: Estimation of GATE $\hat{\Delta}_S$ (BLT vs. SLT) across prespecified clinical subgroups. The horizontal dashed line denotes the average treatment effect.

5. Discussion

In this work, we present a unified framework for the rigorous analysis of heterogeneous treatment effects in the presence of potentially complex confounding structures. Central to this framework is *deepHTL*, a revised deep semiparametric regression method designed to accurately estimate structurally complex treatment effects in clinical practice. To evaluate the presence of treatment effect heterogeneity prior to estimation, we develop a Davies-type test, a kernel-based procedure that ensures valid Type I error control even under intricate nuisance structures. We verify the finite-sample robustness of these methods through extensive simulations and demonstrate their practical utility in a case study on lung transplantation, which reveals clinically meaningful heterogeneity in the efficacy of bilateral versus single-lung transplantation. Although illustrated here in a biomedical setting, the framework applies broadly to disciplines where understanding effect variation is essential.

Our methodology refines and extends existing approaches in several ways. Building on the quasi-oracle R-learner framework of [12], we introduce a revision step that improves the estimation of $\mathbb{E}(Y \mid \boldsymbol{X})$ in situations with large-magnitude treatment effects. We apply the bagged-DNN method of [13] for nuisance and treatment effect estimation, taking advantage of its high capacity and robustness for function approximation when sample size is adequate. Additionally, we address the critical but often overlooked challenge of testing the global existence of heterogeneity prior to estimation. The development of the Davies test in this context represents a novel contribution to the causal inference literature and provides a rigorous procedure for detecting heterogeneity with valid Type I error control.

Despite these advances, several challenges warrant consideration. First, causal identification relies on the standard assumptions of unconfoundedness, positivity, and consistency. As with any observational study, our results are sensitive to unmeasured confounding, and future work should incorporate formal sensitivity analyses to quantify the robustness of findings to hidden bias. Second, the performance of *deepHTL* depends heavily on the accurate estimation of nuisance functions. While the cross-fitting ensemble approach mitigates overfitting, the computational burden of training multiple deep neural networks is substantial. Moreover, when the true treatment effect distribution is highly dispersed or contains extreme values, the centering step based on the ATE may itself become unstable, motivating the development of more robust preliminary estimators. Finally, our current framework addresses static binary treatments with fully observed outcomes. Many medical applications, including the lung transplant study, involve time-to-event outcomes subject to censoring or longitudinal treatment regimes with time-varying confounding. Extending both the Davies-type test and *deepHTL* to handle survival endpoints and dynamic treatment rules deserves further investigation.

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Appendices

Table 3: Baseline patient and donor characteristics used as confounders by transplant type (BLT vs. SLT).

Covariates	**Transplant type**	
	BLT (n = 9758)	**SLT** (n = 4548)
Recipient Age (years), Mean (SD)	53.4 (13.5)	63.5 (7.5)
Recipient BMI, Mean (SD)	24.8 (4.7)	26.3 (4.1)
Lung Allocation Score, Mean (SD)	40.8 (13.5)	40.3 (11.1)
Diabetes, n (%)	1999 (20.5%)	836 (18.4%)
TRR FeV1 %, Mean (SD)	37.2 (20.7)	43.6 (20.4)
TRF FEV1, Mean (SD)	78.2 (24.0)	65.4 (20.1)
Gender Female, n (%)	4081 (41.8%)	1610 (35.4%)
Gender Male, n (%)	5677 (58.2%)	2938 (64.6%)
Recipient Height (cm), Mean (SD)	170.0 (9.9)	170.6 (9.6)
Hispanic, n (%)	684 (7.0%)	305 (6.7%)
O_2 Requirement at Rest, Mean (SD)	5.6 (5.3)	4.7 (4.3)
O_2 Delivered, Mean (SD)	7.0 (6.6)	5.7 (5.2)
Ventilator Use, n (%)	465 (4.8%)	71 (1.6%)
6-Minute Walk Distance, Mean (SD)	773.8 (426.1)	794.6 (411.4)
Donor Age (years), Mean (SD)	34.7 (14.0)	34.9 (13.8)
Black Donor, n (%)	1798 (18.4%)	861 (18.9%)
Donor BMI, Mean (SD)	26.2 (5.6)	26.2 (5.3)
Donor Cigarette Use, n (%)	5499 (56.4%)	3164 (69.6%)
CMV Serology Positive, n (%)	6093 (62.4%)	2793 (61.4%)
Donor TBI (Cause of Death), n (%)	4236 (43.4%)	2006 (44.1%)
Donor Diabetes, n (%)	701 (7.2%)	315 (7.0%)
Distance to TX Center (NM), Mean (SD)	204.0 (246.3)	216.6 (253.3)
Donor Gender Female, n (%)	3857 (39.5%)	1683 (37.0%)
Donor Gender Male, n (%)	5901 (60.5%)	2865 (63.0%)
Donor Height (cm), Mean (SD)	171.7 (10.2)	172.3 (9.9)
Donor PCO_2, Mean (SD)	76.1 (40.1)	75.6 (34.0)
Allo Positive, n (%)	6926 (71.0%)	2949 (64.8%)
HgtR, Mean (SD)	1.0 (0.06)	1.0 (0.05)
Ischemic Time (hours), Mean (SD)	5.6 (1.76)	4.3 (1.52)
Sex Match, n (%)	6708 (68.7%)	3143 (69.1%)
Race Match, n (%)	5494 (56.3%)	2548 (56.0%)

Table 4: Cluster-level mean values of individualized treatment effects and key baseline features.

Cluster	Size	$\hat{\tau}(X)$	LAS	Isch(h)	6MWD(m)	BMI	Age(R)	Age(D)	CMV(%)
C1	2014	25.85	39.50	5.91	861.92	20.15	32.42	31.27	64.00
C2	3420	32.27	33.46	4.62	741.27	22.63	60.67	36.19	69.18
C3	3199	12.53	39.16	5.24	848.53	27.62	61.01	33.91	0.00
C4	1137	8.65	72.79	5.56	304.59	25.58	56.98	35.12	67.02
C5	4426	9.72	39.43	5.11	846.97	27.92	61.17	35.62	100.00

LAS: Lung Allocation Score; Isch(h): ischemic time in hours; 6MWD(m): six-minute walk distance in meters; BMI: recipient body-mass index; Age(R): recipient age; Age(D): donor age; CMV(%): donor CMV seropositive percentage.

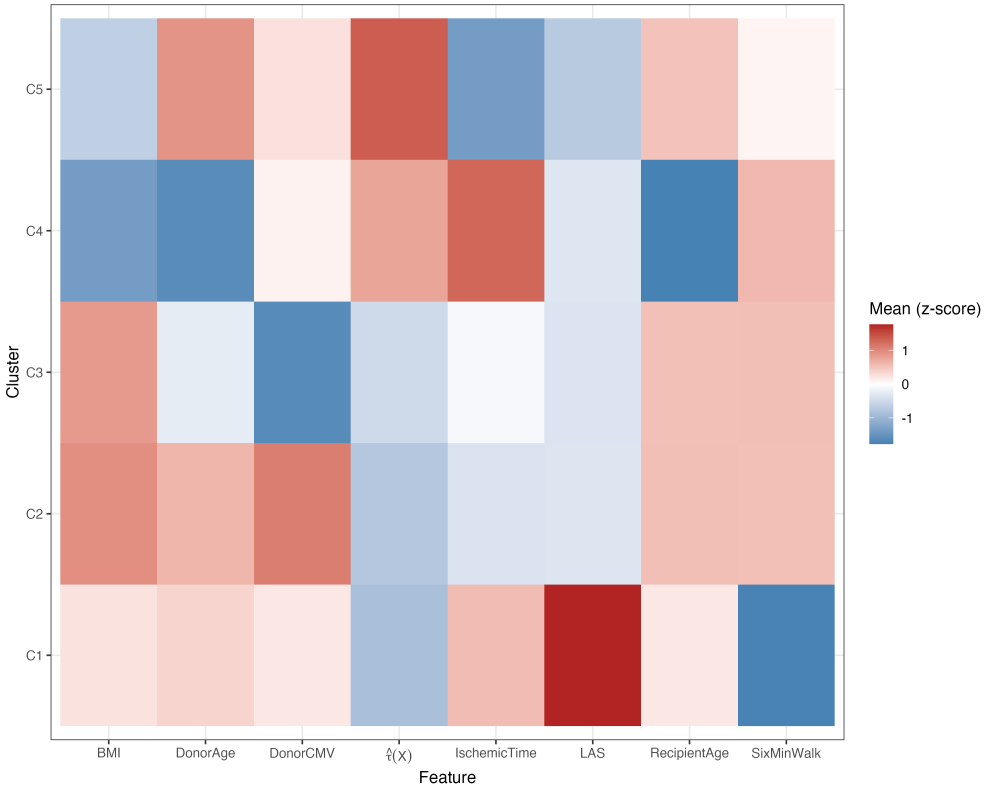

Figure 4: Heatmap of standardized cluster-level feature means derived from k-means (best k = 5 chosen by average silhouette) clustering on the estimated individualized treatment effects $\hat{\tau}(\boldsymbol{X})$ and key clinical covariates. Each cell shows the within-cluster mean (z-scored across features). Red indicates above-average values and blue indicates below-average values. Clusters with higher $\hat{\tau}(\boldsymbol{X})$ are characterized by lower LAS, lower BMI, and younger recipient age, confirming the same heterogeneity pattern observed in the predefined GATE analysis.